

José Fernando Gómez Bedoya

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EDUCATION

2022 - 2026 B.Sc. in Physics at **EIA University** (GPA: 4.5/5.0)
2023 - 2027 B.Sc. in Systems Engineering at **EIA University** (GPA: 4.4/5.0)

RESEARCH EXPERIENCE

Algebraic Structures Research Group, EIA University Jul. 2025 - present

- Studied generalized digroups and related algebraic structures arising in noncommutative and nonassociative settings.
- Investigated subgroup-type decompositions, with emphasis on Sylow-type results in generalized algebraic contexts.
- Explored connections between set-theoretic solutions of the Yang–Baxter equation and algebraic structures such as racks and quandles.

Semiconducting Nanostructures Research Group, EIA University Jan. 2025 - present

- Theoretical study of optoelectronic properties of low-dimensional semiconductor nanostructures.
- Modeling quantum wells and heterostructures under external fields using effective-mass quantum mechanics.
- Analysis of confined electronic states and optical response in asymmetric quantum well systems.
- Results disseminated through oral and poster presentations at national conferences; preprint submitted to proceedings.

ACADEMIC & PROFESSIONAL EXPERIENCE

Teaching Assistant, EIA University Jun. 2024 - Nov. 2024

Provided academic support for the courses of Calculus I-II and Computational Physics I.

Full-Stack Developer, ARUS (Colombia) Nov. 2022 - Apr. 2024

- Designed and maintained modular software systems using Python and JavaScript, with emphasis on robustness, reproducibility, and version control.
- Developed data-driven pipelines and performed computational analysis using scientific Python libraries (scikit-learn, PyTorch), including machine learning methods.
- Implemented containerized and distributed applications using modern frameworks and tooling (Docker, Vue, NestJS, Starlette, among others).

TALKS & POSTERS

Oral Presentation 8–10 Oct. 2025

Asymmetric AlGaAs/GaAs Quantum Wells in a Dielectric Environment: Electric-Field-Tunable Optical Properties, 30th Congress of the Colombian Physical Society.

Poster Presentation 2–3 Oct. 2025

Asymmetric AlGaAs/GaAs Quantum Wells in a Dielectric Environment: Electric-Field-Tunable Optical Properties, Escuela–Congreso “100 años de Revolución Cuántica”, Medellín, Colombia.

PUBLICATIONS

Gómez-Bedoya, José F. and Ricardo L. Restrepo Arango (Dec. 2025). “Asymmetric AlGaAs/GaAs Quantum Wells in a Dielectric Environment: Electric-Field-Tunable Optical Properties”. Preprint. Submitted to the Proceedings of the 30th Congress of the Colombian Physical Society. DOI: [10.5281/zenodo.18062815](https://doi.org/10.5281/zenodo.18062815). URL: <https://doi.org/10.5281/zenodo.18062815>.

SCHOOLS & WORKSHOPS

2nd ConHEP School: School from the Colombian Network on HEP

Nov. 2025

School covering particle physics topics (Standard Model, Beyond Standard Model, Neutrino and Dark Matter physics) and High-Energy Physics computational tools, organized by the Colombian Network on High Energy Physics.

Óptica Cuántica para la Era de la Información

Sep. 2025

School on quantum optics and quantum information, Escuela–Congreso “100 años de Revolución Cuántica”, Medellín, taught by members of the TU Wien Theoretical Quantum Optics group (Atom-institut).

SKILLS

Programming Languages	Python, Julia, C/C++, Fortran, Bash
Scientific & Technical Computing	L ^A T _E X, Mathematica, ROOT
Machine Learning	PyTorch, scikit-learn
Web Development	HTML/CSS/JavaScript, React, Next.js
DevOps & Tooling	Linux, Git, Docker, SQL
HEP Tools (basic)	Pythia, Delphes, Geant4, MadGraph, MadDump, micrOMEGAs

LANGUAGES

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| – Spanish , Native | – French , B1, DELF Certified |
| – English , C1, TOEFL Certified | – German , B1 |